

# How Much Can LLMs Help Solve Mixed-Integer Programs?

Evidence from 112 open MIPLIB instances

An evidence-graded study of where language models helped, where optimization backends still led, and which claims survived independent checking.

|                           |                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------|
| <b>Evidence date</b>      | 18 September 2026 UTC (17 September 2026 PDT)                                                  |
| <b>Published baseline</b> | Commit 89a8abd dated 16 September 2026                                                         |
| <b>Supplements</b>        | Local independent reviews dated 18 September 2026 UTC, not yet committed                       |
| <b>Audience</b>           | Mixed-integer optimization, exact algorithms, automated reasoning, and AI-assisted mathematics |

# Contents

|                                                                                           |           |
|-------------------------------------------------------------------------------------------|-----------|
| <b>Executive summary</b>                                                                  | <b>1</b>  |
| <b>1 Research question: what counts as help?</b>                                          | <b>1</b>  |
| <b>2 Acceptance rule: discovery is not proof</b>                                          | <b>2</b>  |
| <b>3 Division of labor and the LLM-assisted research loop</b>                             | <b>3</b>  |
| <b>4 How much help was observed?</b>                                                      | <b>4</b>  |
| 4.1 Coverage . . . . .                                                                    | 4         |
| 4.2 Primal and dual progress . . . . .                                                    | 5         |
| 4.3 Verification performed for this report . . . . .                                      | 5         |
| <b>5 Closest available comparison: a mixed result</b>                                     | <b>5</b>  |
| <b>6 Four modes of LLM contribution</b>                                                   | <b>6</b>  |
| 6.1 Invariant reconstruction: parity in <code>allcolor58</code> . . . . .                 | 6         |
| 6.2 State-space redesign: tree dynamic programming in <code>dfn-bwin-DBE</code> . . . . . | 6         |
| 6.3 Semantic recovery: interval orders in <code>fhnw-binschedule2</code> . . . . .        | 7         |
| 6.4 Proof-backend orchestration: SAT in <code>bppc6-06</code> . . . . .                   | 7         |
| <b>7 Where LLM assistance failed or misled</b>                                            | <b>7</b>  |
| <b>8 Remaining research frontier</b>                                                      | <b>8</b>  |
| <b>9 Threats to validity</b>                                                              | <b>8</b>  |
| 9.1 Selection and budget bias . . . . .                                                   | 8         |
| 9.2 Evidence heterogeneity . . . . .                                                      | 9         |
| 9.3 Official status lag . . . . .                                                         | 9         |
| 9.4 External and missing artifacts . . . . .                                              | 9         |
| 9.5 Reproducibility details . . . . .                                                     | 9         |
| 9.6 Source synchronization . . . . .                                                      | 9         |
| <b>10 Reproducibility contract</b>                                                        | <b>9</b>  |
| <b>11 Conclusion</b>                                                                      | <b>10</b> |
| <b>A Reconciled repository supported global conclusions</b>                               | <b>11</b> |
| <b>References</b>                                                                         | <b>12</b> |

## Executive summary

**Short answer.** Language models helped materially as research copilots, especially in semantic recovery, state-space redesign, experiment construction, falsification, and proof orchestration. They did not replace mixed-integer solvers, SAT engines, exact enumeration, or deterministic checking. The durable contribution of the MIPLIB\_awesome project is the loop connecting LLM-generated hypotheses to independently checkable mathematical objects.

On the fixed 4 September 2026 snapshot of 217 officially open MIPLIB instances, the LLM-assisted campaign studied 112 current open instances and retained one historical calibration instance. The reconciled working tree supports 30 global conclusions among the 112 studied instances: 28 optimal values and two infeasibility results. Eighty two studied instances remain globally unresolved. These numbers describe repository evidence, not updates to the official MIPLIB status pages, and they do not estimate a success probability because instance selection and research budgets were nonuniform [1].

The repository also records 21 improvements over existing MIPLIB v36 primal values, four first finite incumbents, and 41 improved lower bounds relative to a frozen historical COPT 10 hour table. These are different, overlapping outcomes of the campaign. They are not 96 distinct wins, the primal and dual counts do not share a baseline, and the lower-bound comparison is not a same-hardware or same-budget solver benchmark.

The central findings are:

1. The LLM was most useful when it helped replace a large anonymous formulation by a smaller object: a tree, interval order, parity argument, finite cutoff, or proof-producing SAT instance.
2. Construction and proof remain different pipelines. A strong feasible solution does not prove a lower bound, and a lower bound does not supply a witness.
3. Solvers search and bound; deterministic checkers decide what survives. Reduced-model optimality, local infeasibility, and unchecked SAT unsatisfiability remain scoped statements.
4. The strongest direct comparison is mixed. In a nonrandomized 20-instance study, the structure arm recorded better selected primals and two exact closures, while the comparison arm more often retained stronger numerical dual bounds.
5. The campaign counts establish research output, not sole LLM causation or a universal success rate.

## 1 Research question: what counts as help?

The capability question has four outcome dimensions: strict primal progress, a valid global dual improvement, a supported global conclusion, and proof portability. An LLM can contribute by interpreting the matrix, proposing a representation, designing a falsification test, or generating a checker. The claim itself counts only after the appropriate solver, exact computation, original-MPS audit, or proof replay succeeds.

For a minimization problem, the project uses the basic proof obligation  $\mathbf{LB} \leq \mathbf{OPT} \leq \mathbf{UB}$ , with global optimality established only when a valid lower bound and a strictly feasible upper bound coincide.

The phrase **open instance** is frozen to the MIPLIB open list downloaded on 4 September 2026 [13]. It means the official record at that snapshot did not contain a global closure. It does not mean that no later local computation, external construction, or published theorem exists. The project therefore separates three statuses:

- the official status in the frozen MIPLIB snapshot;

- the conclusion supported by the repository evidence; and
- whether that conclusion has a portable, independently replayable proof.

The numerical baseline for this report is the 16 September 2026 presentation [4] and the 14 September dual bound audit [3], reconciled with the two 18 September working tree reviews. The 14 September audit upgrades the five `fhnw-schedule-pair` instances from solver based closures to exact branch and dual certificates and corrects several terminal decimal digits.

The uncommitted 18 September first five review is treated as a supplement [11]. It hardens the `assign1-10-4` and `fhnw-binschedule0` checks and fills a proof scope gap in `cvrpp-n16k8vrpi` by covering every nonempty route count from one through fifteen. Because that package is not yet in the published commit, this report marks the CVRPP classification as pending integration even though the supplemental audit again obtains value 450.

The companion review of rows 6 through 20 [12] confirms all fifteen advertised lower bounds but withdraws two exact-decimal optimality claims. For `genus-g31-8` and `genus-sym-g31-8`, the native graph optimum and original-MPS lower bound are both  $-23$ , while the saved objective  $-23$  witnesses violate six and seven literal decimal MPS rows, respectively, by  $10^{-15}$ . Without a zero-tolerance witness, those two instances remain unresolved in this report.

## 2 Acceptance rule: discovery is not proof

The project deliberately avoids one undifferentiated solved label. Its evidence classes answer different questions.

| Evidence class                                                    | Opt.      | Infeas.  | Total     | What is checked                                                                                |
|-------------------------------------------------------------------|-----------|----------|-----------|------------------------------------------------------------------------------------------------|
| Self-contained portable exact certificate                         | 17        | 1        | 18*       | Original model mapping, exact bound, witness, and solver-free replay                           |
| Checked exact result with large trace stored outside Git by hash  | 1         | 0        | 1         | Compact mapping and logs are local; the 2.65 GiB proof trace must be obtained or regenerated   |
| Specialized exact exhaustive computation without standalone trace | 3         | 0        | 3         | Exact program and output are archived, but not a small independent proof object                |
| Exact model crosswalk plus published result                       | 2         | 1        | 3         | Local mapping is checked; the external theorem or computation is accepted rather than replayed |
| Floating-point solver closure                                     | 3         | 0        | 3         | Strict primal audit plus commercial global lower bound; no portable exact lower bound trace    |
| Mixed database and computational argument                         | 2         | 0        | 2         | Exact projection is combined with external benchmark values and commercial exclusions          |
| <b>Total</b>                                                      | <b>28</b> | <b>2</b> | <b>30</b> |                                                                                                |

**Table 1:** Reconciled working tree evidence classification for the 30 repository supported global conclusions.

The 18 portable exact conclusions include the five exact FHNW scheduling certificates completed in the 14 September audit. The table gives a reconciled working tree classification, not a commit only classification. The asterisk marks one material qualification: the committed CVRPP proof fixed eight routes even though the compiled model permits up to fifteen vehicle slots. The uncommitted supplement closes that scope gap; until it is integrated, the portable classification should carry this audit note.

The exact exhaustive class has three members after the two g31 conclusions were removed. The three literature transfers rely on exact local crosswalks to externally published results [16][17][18].

This evidence model also defines what weaker artifacts do not prove:

- A feasible candidate accepted at a loose tolerance is not yet a strict incumbent.
- **OPTIMAL** on a projection proves the original problem only if the projection direction and lifting argument are correct.
- **INFEASIBLE** in a fixed neighborhood excludes only that neighborhood.
- A time limit is neither feasibility nor infeasibility evidence.
- SAT or pseudo Boolean **UNKNOWN** is not an unsatisfiability result.
- **UNSAT** is not portable without a checked encoding implication and a replayed proof trace.
- A paper value transfers only after the exact instance identity and model crosswalk are established.

### 3 Division of labor and the LLM-assisted research loop

The campaign assigns different responsibilities to the language model, computational backends, and independent checks. This separation prevents fluent conjectures or successful restricted runs from being promoted into global claims.

| Component               | Primary role                                                                                                            | What it does not establish by itself                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Language model          | Interprets structure, proposes reductions and experiments, writes extractors and checkers, and coordinates proof routes | Feasibility, a global bound, optimality, or correctness of generated code               |
| Solver or exact backend | Searches for incumbents and bounds; executes dynamic programs, enumeration, or SAT encodings                            | Validity outside the solved formulation, tolerance, neighborhood, or time-limited scope |
| Independent checker     | Audits the original MPS, mappings, witnesses, proof traces, and exact arithmetic                                        | Novelty or causal attribution to the LLM                                                |

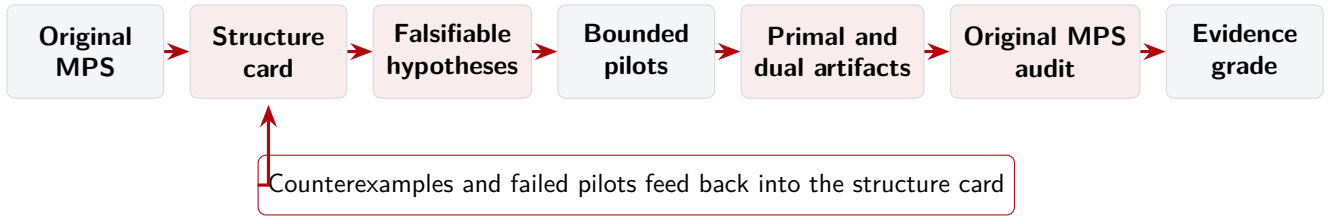
*Table 2: The operational division of labor used throughout the campaign.*

The recurring protocol can be summarized as follows.

1. **Freeze the target claim.** Decide whether the goal is a primal improvement, a dual bound, feasibility, infeasibility, or global closure.
2. **Read the untouched MPS.** Record variable domains, objective support, row senses, coefficient patterns, sparsity, repeated columns, numerical scales, and candidate graph or block structure.
3. **Recover semantics and provenance.** Identify the underlying problem family, sibling formulations, public solutions, and relevant theorems without treating names or metadata as proof.
4. **Form falsifiable structural hypotheses.** Each proposed reduction or inequality is paired with a small counterexample search or exact consistency check.
5. **Run bounded pilots.** Experiments have explicit scope and stop conditions. A failed pilot updates the model of the problem rather than disappearing from the record.

6. **Separate the two pipelines.** The primal pipeline constructs and repairs witnesses. The dual pipeline develops relaxations, counting arguments, dynamic programs, proof trees, or cutoff refutations.
7. **Return to the original MPS.** Reconstruct every variable needed for a full solution, verify every row, bound, and integrality condition, and audit objective constants and signs.
8. **Grade the evidence.** State whether the result is exact, exhaustive, theorem dependent, solver numerical, local, or conditional.
9. **Stop or change representation.** Repeated null steps, exhausted neighborhoods, invalid cuts, or incomplete proof traces are signals to change the state space rather than lengthen the same run.

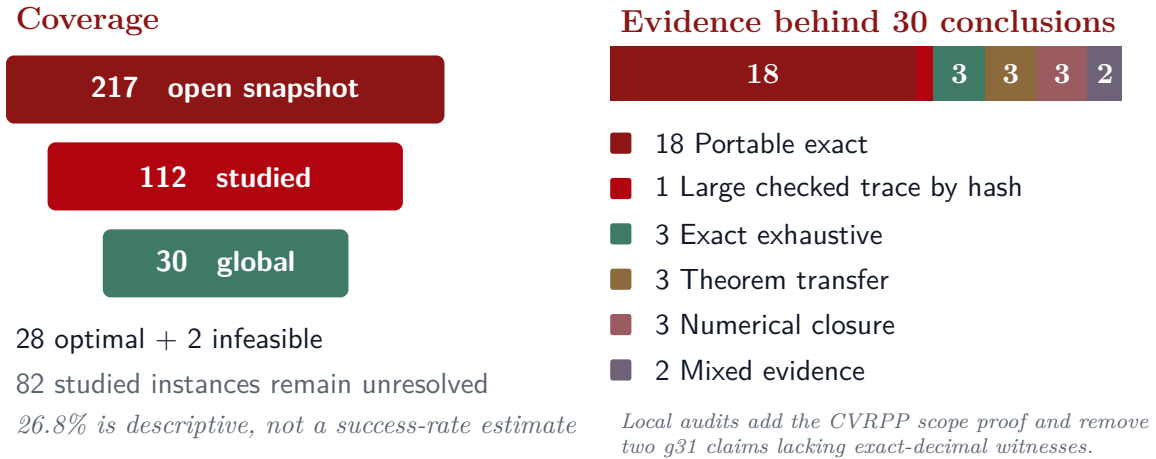
The key design principle is that expression complexity and combinatorial complexity differ. A model with hundreds of thousands of columns can be controlled by a few hundred selections, a path cover in an interval graph, a small separator state, a finite target cutoff, or a handful of coupling rows. The research task is to find the smallest representation that preserves the direction of inference needed for the claim.



*Figure 1: LLM-assisted research workflow with explicit falsification and return to the original MPS.*

## 4 How much help was observed?

### 4.1 Coverage



*Figure 2: Frozen campaign coverage and evidence mix. Evidence classes preserve materially different dependencies.*

| Metric                                        | Count        | Interpretation                                                 |
|-----------------------------------------------|--------------|----------------------------------------------------------------|
| Open instances in frozen 4 September snapshot | 217          | Historical universe, not a live status query                   |
| Current open instances studied                | 112          | Plus one historical calibration instance, <b>rmine14</b>       |
| Repository supported global conclusions       | 30           | 28 optimal and 2 infeasible                                    |
| Studied but globally unresolved               | 82           | Includes useful primal, dual, structural, and negative results |
| Closure fraction within studied set           | 26.8 percent | Descriptive only; selection and budgets were nonuniform        |

*Table 3: Frozen campaign coverage and interpretation.*

## 4.2 Primal and dual progress

The repository counts 21 numerical improvements over the official MIPLIB v36 solution baseline [14] and four first finite incumbents, for 25 primal advances. The count includes two **fastxgemm** improvements derived from an attributed external construction; the independent project contribution is the truncation, mapping, and strict verification, not authorship of the parent construction. The published R23 construction is reproduced but not counted as a project primal improvement.

The 41 lower-bound advantages compare repository bounds with displayed values in a frozen historical COPT 10 hour table [15]. They mix exact certificates, theorem transfers, commercial solver bounds, and numerical evidence. They therefore measure archived bound progress, not solver speed. Two notable exact but still nonclosing bounds are **graphdraw-mainerd**  $\geq 31,882$  and **graphdraw-opmanager**  $\geq 58,950$ .

## 4.3 Verification performed for this report

Two solver free replay suites were executed in temporary directories during preparation of this report.

- The 14 September portable suite completed all 15 tasks with exit status zero. It replayed five FHNW scheduling certificates, **ramos3**, **sorrell17**, two **graphdraw** certificates, the exact large route CVRPA bound, negative controls, and both CVRP projection and primal checks. No commercial optimizer was called.
- The frozen ten-instance publication verified all 259 file hashes and completed all ten requested checks. It replayed compact combinatorial certificates, zero-tolerance witnesses, genus catalogues, an exact branch tree, and an LRAT proof. It launched no optimization search.

These replays do not cover the external theorem computations, the absent 2.65 GiB BPPC proof trace, specialized exhaustive computations lacking standalone traces, or historical commercial solver searches.

## 5 Closest available comparison: a mixed result

A separate historical study compared a structure research skill with a run that did not invoke it on twenty instances [9]. Under its absolute  $1e-7$  selected-primal rule, the structure arm recorded better values on eleven instances and tied on nine. It recorded two exact global closures, versus none in the comparison arm. Six selected values were tolerance indexed rather than strict incumbents; against strict public baselines, the improvement counts were four for the structure arm and two for the comparison arm. The comparison arm retained stronger dual bounds on twelve of twenty instances and smaller common gaps on eleven of twenty. The structure arm’s two closures, **ns1456591** and **neos-3682128-sandon**, remain in

that separate comparison archive and are not integrated into the canonical 112-instance, 30-conclusion inventory.

This is observational evidence, not a causal estimate. The runs were not randomized, blinded, or equal budget; machines, starts, solver states, and stopping decisions differed. The defensible conclusion is operational:

- structure-first work is valuable for semantic recovery, new representations, primal construction, certificate design, and original-model audit;
- a protected generic solver budget remains valuable for numerical dual progress when the structural route does not close the instance.

The comparison therefore supports a hybrid research design rather than replacing branch and bound with a language model.

## 6 Four modes of LLM contribution

The following cases show four distinct forms of help. In each case, the language model’s contribution is separated from the computational backend, the verified outcome, and the remaining limitation. They are examples of an LLM-assisted process, not claims that the model alone discovered every theorem, construction, or benchmark value.

### 6.1 Invariant reconstruction: parity in allcolor58

allcolor58 has 84,376 variables and 197,154 rows, but its decisive lower bound is a parity argument. Every legal package has even total capacity, and assignment counts are integral, so each store receives an even total quantity. Exactly 42 stores have odd total demand. Every such store must incur at least one unit of overstock or understock cost, proving a global lower bound of 42.

The value 42 and a construction seed were already present in external or literature information. The repository contribution is therefore not a claim of first discovery: it reconstructs the parity lower bound, produces a replayable full witness from a compact search, and subjects that witness to exact original model audits [5].

The primal side is separate. A factorized search constructs fourteen package configurations whose allocations give exactly one unit of overstock at each odd demand store and no other deviation. Two independent exact audits verify all variables and all original rows with zero violation. Thus the parity lower bound and the strict witness meet at 42.

This case also records an important historical correction. Earlier objective zero candidates failed independent verification. They remain in the archive as failed experiments rather than being rewritten as progress. The eventual proof did not come from more aggressive tuning of those candidates; it came from changing the representation.

### 6.2 State-space redesign: tree dynamic programming in dfn-bwin-DBE

The network design model has ten nodes and positive demand for every ordered pair. Any feasible installed support must therefore be connected. A connected support with nine physical edges is a spanning tree. Any support with ten or more edges costs at least the sum of the ten cheapest positive opening costs, 81,222.40, already greater than the candidate value 73,623.79.

It remains to optimize over spanning trees. Removing a tree edge creates a cut; unique tree paths force all cross cut demand through that edge. The two directional totals determine the minimum feasible

capacity option and its exact cost. Two independent subset dynamic programs enumerate the rooted tree decompositions and obtain 7,362,379 cents for every root. The generated tree solution passes an exact original model audit, completing the proof of 73,623.79 [6].

The conceptual reduction is global, not heuristic. The ten-edge cost bound excludes every denser support, while the dynamic program covers every tree support. The proof is not a claim that a good solution happened to be a tree.

### 6.3 Semantic recovery: interval orders in fhnw-binschedule2

The original MPS is a six machine makespan formulation with 497 items, 114 ordered positions, 1,614 local OR variables, 1,908 selector windows, and 228 difference indicators. Seventy six items can use only the two 39 position endpoint machines.

Recovering the selector windows exposes an interval order. A set of 35 intervals all contains the same interior point, so its members are pairwise incomparable. If endpoint machine  $k$  receives  $n_k$  mandatory items and activates  $D_k$  differences, its sequence can cover at most  $40 - n_k + D_k$  chains. Summing over both endpoints forces at least 31 active differences.

The mandatory item costs total 4,546 and every difference costs 10, so the two endpoint loads sum to at least 4,856. Endpoint loads lie on an even integer lattice. If the makespan were below 2,428, both endpoint loads would be at most 2,426 and their sum at most 4,852, a contradiction. The public solution attains 2,428 and passes exact checking of all 6,503 rows [7].

The proof combines two structures that a generic relaxation largely obscures: an interval antichain controls the number of breaks, and a lattice restriction removes the remaining numerical slack.

### 6.4 Proof-backend orchestration: SAT in bppc6-06

bppc6-06 chooses one start position for each of twenty items on a horizon of 300 and minimizes the maximum profile height. A known placement has height 208. The lower bound task is therefore the finite decision question: can height 207 occur?

An exact subset enumeration finds nonnegative integer weights for the items such that every subset with physical height at most 207 has weight at most four. The total weighted length is exactly four times the horizon, so every positive width profile segment in a hypothetical height 207 placement must have weight exactly four. This creates a much stronger Boolean encoding than the raw height constraints alone.

The resulting CNF has 414,391 Boolean variables and 1,541,322 clauses. CaDiCaL reports unsatisfiable, drat-trim verifies the proof, and a separate checker replays 1,001,496 LRAT lemmas through the final empty clause. Two earlier incomplete proof logs are explicitly rejected. The exact 208 placement passes the original MPS audit, so the checked cutoff refutation and witness establish the optimum [8].

The limitation is equally explicit: the complete CNF, DRAT, and LRAT artifacts total about 2.65 GiB and are stored outside Git by byte size and hash. The repository contains the encoder, compact mapping, manifests, and replay summaries, but an independent reviewer must obtain or regenerate the large files for end-to-end replay.

## 7 Where LLM assistance failed or misled

The archive is unusually useful because it records incorrect turns and their scope.

An early FHNW scheduling route interpreted a latest start bound as a completion deadline. That missing duration term produced invalid energy cuts and apparent zero gaps. Original model counterexamples forced the dual claims to be withdrawn. The final five proofs use safe release suffix relaxations, complete

binary branch trees, exact rational leaf duals, and full original MPS witnesses. The correction changed both the mathematics and the evidence grade.

Other recurring failure modes include:

- parsers that miss indented RHS records;
- row orientation errors in pseudo Boolean encodings;
- reductions that are valid as relaxations but are mistakenly treated as liftable equivalences;
- time-limited SAT or MILP runs written as if they had excluded a target;
- a fixed support or local Hamming ball confused with the full feasible region;
- floating-point solutions whose apparent improvement vanishes after exact row and integrality checks;
- proof generation logs truncated before the final certificate is flushed.

The response is a persistent experimental ledger. Each result states the precise subproblem, target, time limit, terminal status, mapping direction, and verification performed. A failed experiment is valuable when it rules out a mechanism or identifies the minimum scale of the next move. It is not evidence of global impossibility.

## 8 Remaining research frontier

The 82 unresolved studied instances contain more than unfinished solver runs. Many already have exact intervals, verified reductions, completed local exclusions, or proof kernels. The frozen continuation ranking [10] predates the two g31 qualifications and therefore covers 80 of the 82 current unresolved instances. It argues that the best next target is not necessarily the smallest displayed gap. Priority should instead go to an instance already reduced to a small auditable decision problem with a plausible certificate route.

The leading continuation portfolio in the 9 September snapshot is:

1. `graph40-20-1rand`, where a sibling SAT and LRAT template suggests a direct 16 cut refutation after graph identity is certified;
2. `genus-sym-g62-2`, where only a small recorded frontier remains in an exact face search;
3. `genus-g61-25`, where the next even face target is finite and checkpointed;
4. `ns1828997`, where a very small strict gap and objective lattice make a cutoff proof plausible after exact reproduction.

The ranking is a research allocation heuristic, not a calibrated probability of solution. Each continuation needs a stage gate that confirms the required inputs, mapping, and proof backend before a long run.

## 9 Threats to validity

### 9.1 Selection and budget bias

The 112 studied instances were not sampled randomly, and the project preferentially pursued visible structure, tractable certificates, and promising siblings. Research budgets varied. The ratio 30 divided by 112 is therefore a historical closure fraction, not a general success rate.

## 9.2 Evidence heterogeneity

Exact certificates, specialized exhaustive programs, published theorems, commercial zero gap logs, and mixed database arguments do not provide the same portability. Aggregate counts must preserve their evidence mix.

## 9.3 Official status lag

Repository conclusions may postdate the frozen MIPLIB solution file or remain unreviewed upstream. This report never treats a repository closure as an official status change.

## 9.4 External and missing artifacts

The large BPPC trace is external by hash; literature computations were not replayed; three exact exhaustive results lack compact independent traces; three closures are floating-point solver closures; two mixed CVRP conclusions additionally depend on commercial route-count exclusions; and seventeen instance folders retain provenance rather than a local MPS copy. The two 18 September reconciliation audits are uncommitted source-repository working tree supplements and are not bundled with LLM4MIP, so public readers cannot replay them until they are published or copied into a stable evidence package.

## 9.5 Reproducibility details

Some scripts require Python 3.10 or later. The shared exact solution checker prints violation counts but does not always fail through its process status, so reviewers must inspect the recorded counts. The `minutedispatchstrategy` archive retains a conflict with an old unavailable PDF primal value, and the `cmflsp40-36-2-10` numerical result lacks its original global runner.

## 9.6 Source synchronization

The root README at published commit `89a8abd` of the source `MIPLIB_openproblem` repository contains stale evidence counts and stale FHNW labels. The 14 September audit and 16 September presentation are the later authorities for evidence classes and exact decimals. The report preserves this precedence instead of averaging conflicting summaries.

# 10 Reproducibility contract

A publication quality result should archive the following components.

1. The exact original MPS or a stable official URL, byte count, and cryptographic hash.
2. The objective sense, objective offset, variable domains, row senses, and numerical scaling.
3. A proof of the direction from every original feasible point to the reduced object used for the bound.
4. A lifting or reconstruction procedure when reduced feasible points are used to build original witnesses.
5. A complete original model solution file with exact or explicitly bounded tolerance checks of every row, bound, and integrality condition.
6. A certificate whose semantics are independent of the search program whenever practical.
7. An independent checker, negative controls, software version, and replay command.
8. Hashes for all frozen source, certificate, and checker files.

9. An evidence label and an explicit list of what was not replayed.
10. A distinction between the repository result and the official benchmark status.

The archive already follows much of this contract, but the supplemental audits show why it matters. A correct value can coexist with a checker scope gap. Independent parsing and a second proof route convert that value into a more trustworthy result.

## 11 Conclusion

How much can LLMs help solve MIPs? This campaign shows material assistance: the archive records 30 supported global conclusions, 25 primal advances, and 41 historical lower-bound advantages across 112 studied open instances. The closest 20-instance comparison also records an operational advantage in selected primals and exact closures. None of those counts isolates the LLM’s marginal contribution, estimates a randomized causal effect, defines a universal success rate, or shows that the language model can replace an optimizer.

The defensible conclusion is narrower and more useful. LLMs can materially expand the search for semantics, representations, falsification tests, and certificate routes. Generic solvers remain essential for numerical search and bounds; exact scripts, original-MPS audits, and proof systems enforce the boundary between a plausible idea and a proved statement.

The strongest examples are changes of state space rather than parameter searches: even capacity becomes parity, network design becomes a tree dynamic program, scheduling becomes an interval chain cover, and a narrow packing cutoff becomes a proof-producing SAT instance. The reusable scientific asset is the loop that connects those representations back to the original MPS and assigns each conclusion the evidence grade it actually earns. The most valuable AI contribution is often a smaller question whose answer can be independently checked.

## A Reconciled repository supported global conclusions

Abbreviations: PE portable exact certificate; HP checked exact proof with large trace external by hash; EX specialized exact exhaustive computation; LT exact crosswalk plus published theorem or result; NS numerical solver closure; MX mixed database and computational evidence.

*Table 4: Reconciled repository supported global conclusions and evidence grades.*

| Instance               | Conclusion                                              | Grade           |
|------------------------|---------------------------------------------------------|-----------------|
| assign1-10-4           | OPT = 422                                               | PE              |
| cvrpp-n16k8vrpi        | OPT = 450                                               | PE*             |
| dfn-bwin-DBE           | OPT = 73,623.79                                         | PE              |
| fhnw-binschedule0      | OPT = 15,958                                            | PE              |
| fhnw-binschedule1      | OPT = 55,158                                            | PE              |
| fhnw-binschedule2      | OPT = 2,428                                             | PE              |
| fhnw-schedule-paira100 | OPT = -15.113116512593401992                            | PE              |
| fhnw-schedule-paira200 | OPT = -19.369961204645205810                            | PE              |
| fhnw-schedule-pairb200 | OPT = -19.369961204645205810                            | PE              |
| fhnw-schedule-paira400 | OPT = -36.276966742368865136                            | PE              |
| fhnw-schedule-pairb400 | OPT = -36.276966742368865136                            | PE              |
| neos-3009394-lami      | OPT = 5.5                                               | PE              |
| neos-2629914-sudost    | OPT = 48,180                                            | PE              |
| allcolor58             | OPT = 42                                                | PE              |
| graph40-80-1rand       | OPT = -7                                                | PE              |
| graph40-40-1rand       | OPT = -9                                                | PE              |
| neos-4355351-swalm     | OPT = 33.45763                                          | PE              |
| bppc6-06               | OPT = 208                                               | HP              |
| neos-4360552-sangro    | OPT = -8                                                | EX              |
| ns1631475              | OPT = 11,100                                            | EX              |
| genus-sym-grafo5708-48 | OPT = -21                                               | EX              |
| circ10-3               | OPT = 242                                               | LT              |
| a2864-99blp            | OPT = -257                                              | LT              |
| neos-3594536-henty     | OPT approximately 401,092                               | NS              |
| neos-2978205-isar      | OPT approximately -11.925808687                         | NS              |
| minutedispatchstrategy | repository closure approximately<br>3,109.9034777596576 | NS <sup>†</sup> |
| cvrpb-n45k5vrpi        | OPT = 751                                               | MX              |
| cvrpa-n64k9vrpi        | OPT = 1,401                                             | MX              |
| fhnw-binpack4-58       | infeasible                                              | PE              |
| pythago7825            | infeasible                                              | LT              |

*\*CVRPP's portable grade uses the 18 September local scope supplement, which is pending integration into the published commit. <sup>†</sup>The minutedispatchstrategy closure retains an unresolved discrepancy with an older unavailable-PDF primal value. The two g31 models are excluded because their saved  $-23$  points fail literal exact-decimal MPS feasibility by  $10^{-15}$ , although their lower bounds and native graph optima are confirmed.*

## References

1. Repository overview at commit 89a8abd
2. Structure driven MILP research strategy
3. Twenty instance dual bound audit
4. 16 September presentation source
5. Allcolor58 exact certificate and attribution note
6. DFN exact dynamic programming proof
7. FHNW binschedule2 exact interval certificate
8. BPPC6 06 checked SAT proof package
9. Historical structure workflow comparison
10. Ranking of 80 unresolved studied instances
11. Local first five independent review, uncommitted source-repository working tree supplement.  
MIPLIB\_openproblem/docs/first-five-dual-bound-review-2026-09-18/README.md
12. Local rows 6 through 20 independent review, uncommitted source-repository working tree supplement.  
MIPLIB\_openproblem/docs/rows-06-20-dual-bound-review-2026-09-18/README.md
13. MIPLIB open instance list
14. Official MIPLIB 2017 v36 solution file
15. Frozen historical COPT 10 hour comparison table
16. Published CIRC10 exhaustive result used by the circ10-3 crosswalk
17. Published finite geometry result used by the a2864-99blp crosswalk
18. Published Boolean Pythagorean triples result used by the pythago7825 crosswalk